

SUMMARY

This video explains the mathematical structure of Partial Order Relations (POR) and their graphical representation through Hasse Diagrams (Poset Diagrams). A Hasse Diagram simplifies a partially ordered set by removing redundant elements such as reflexive self-loops and transitive edges, while preserving the essential ordering structure through vertical positioning.

Key insights include:

A relation must be reflexive, antisymmetric, and transitive to qualify as a partial order.

The maximum size of a partial order relation on a set of size n is $(n^2 + n) / 2$.

Hasse Diagrams remove self-loops and transitive edges.

Totally Ordered Sets always appear as linear chains in Hasse Diagrams.

Divisor sets D_n frequently appear in Poset construction problems.

I. FOUNDATIONS OF PARTIAL ORDER RELATIONS

A relation R on set A is a Partial Order Relation if it satisfies:

1. Reflexivity
For every $a \in A$, $(a, a) \in R$
2. Antisymmetry
If $(a, b) \in R$ and $(b, a) \in R$, then $a = b$
3. Transitivity
If $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$

Cardinality Extremes

Type of POR	Description
Smallest POR	Diagonal relation $\Delta A = \{(a, a)\}$
Largest POR	Contains all diagonals + exactly half of non-diagonals

Largest POR Size Formula

For $|A| = n$:

Maximum cardinality = $(n^2 + n) / 2$

Explanation

Total possible ordered pairs = n^2

Diagonal pairs = n

Non-diagonal pairs = $n^2 - n$

Half of non-diagonals allowed under antisymmetry

Thus:

$$n + (n^2 - n)/2 = (n^2 + n)/2$$

II. POSETS AND COMPARABILITY

A Poset is an ordered pair (A, R) where R is a partial order on A .

Comparability

Two elements a and b are comparable if:

$$(a, b) \in R \text{ or } (b, a) \in R$$

In a general Poset, some element pairs may be incomparable.

Totally Ordered Set (Toset)

A Toset is a Poset in which every pair of elements is comparable.

Property	Description
General Poset	Some pairs may be incomparable
Totally Ordered Set	Every pair is comparable
Example Toset	$(\mathbb{R}, \leq)$

III. HASSE DIAGRAM CONSTRUCTION RULES

A Hasse Diagram is a simplified graphical representation of a Poset.

Construction Rules

1. One vertex per element of A
2. No self-loops
Reflexivity is implied
3. Remove transitive edges
If $a \rightarrow x$ and $x \rightarrow b$ exist, do not draw $a \rightarrow b$
4. Vertical positioning implies direction
If aRb , place a below b

5. Same level placement
Non-comparable elements may be placed at same horizontal level
6. Draw edge only if relation is direct
No intermediate element between them

IV. MATHEMATICAL APPLICATIONS AND EXAMPLES

Divisor Set Notation

D_n = set of all positive divisors of integer n

Examples:

$$D_6 = \{1, 2, 3, 6\}$$

$$D_9 = \{1, 3, 9\}$$

$$D_{12} = \{1, 2, 3, 4, 6, 12\}$$

Example Structures

Set and Relation	Diagram Description
$\{-2, 0, 3, 7, 9\}$ under $\leq$	Single vertical chain: $-2 \rightarrow 0 \rightarrow 3 \rightarrow 7 \rightarrow 9$
D_6 under divides	1 at bottom $\rightarrow$ branches to 2 and 3 $\rightarrow$ both connect to 6
$\{3, 5, 6, 15\}$ under divides	3 and 5 at bottom; $3 \rightarrow 6$ and 15 ; $5 \rightarrow 15$
$\{2, 3, 6, 12\}$ under divides	2 and 3 $\rightarrow 6 \rightarrow 12$ (diamond-shaped structure)

Interpretation Notes

Divides relation creates layered structure based on factor hierarchy.

Totally ordered relations always produce a straight vertical chain.

Branching indicates partial ordering with incomparable elements.